How the Internet Works

Networking Foundations — End of Module Assessment | Video Script (under 3 minutes)

Introduction

Hello everyone. Today, I am going to explain how the internet works.

1. How Computers Are Connected

The internet is a global network of interconnected computers and devices. Devices connect through Wi-Fi, Ethernet cables, routers, switches, fibre-optic cables, and wireless networks. These connections allow data to travel between devices across the world.

2. IP Addresses, Routing, and Subnetting

Every device communicating over the internet uses an IP address to identify its network interface. Devices may receive IP addresses automatically through DHCP or have them configured manually. Routers use destination IP addresses and routing tables to forward packets toward their destinations. Subnetting divides IP networks into smaller networks and helps organize IP address allocation.

3. TCP and UDP

TCP and UDP are transport-layer protocols. TCP establishes a connection and provides reliable, ordered delivery of data, making it suitable for web pages and file transfers. UDP sends data without establishing a connection and has less overhead, making it useful for online gaming, live streaming, and voice calls where speed matters.

4. What Happens When We Type a URL?

When we type a URL into a browser, DNS translates the domain name into an IP address. The browser then connects to the server using the appropriate port, commonly port 443 for HTTPS. The browser sends an HTTP request, and the server responds with the requested content. The browser processes the HTML, CSS, JavaScript, and other resources to display the website. HTTPS also encrypts data exchanged between the browser and server.

5. Packet Switching and Networking Models

Data travels across the internet in small units called packets. These packets may travel through different routers and are reassembled at the destination. This method is called packet switching.

The TCP/IP model describes internet communication using four layers: Application, Transport, Internet, and Network Access. The OSI model has seven layers: Physical, Data Link, Network, Transport, Session, Presentation, and Application. The OSI model is useful for understanding networking concepts and troubleshooting problems, while the TCP/IP model describes the protocols used in real-world internet communication.

For example, a cable problem can be investigated at the Physical layer, an IP addressing problem at the Network layer, and an HTTP problem at the Application layer.

Conclusion

In conclusion, the internet works through connected devices, IP addressing, routing, packet switching, DNS, and communication protocols such as TCP and UDP. Thank you!